Nyaya.AI – Democratizing Legal Knowledge and Services through AI

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***Abstract*—Access to justice sits in the Indian Constitution as a basic right - yet crores of people still do not know what the law gives them. Courts and laws exist in plenty, but a villager meets four walls - hard words in Acts, distant courthouses, fees that empty the pocket and a tongue the judge does not speak. Countryside plus small towns feel the pinch hardest, because few there read law books. Nyaya.AI is a computer site that speaks plain words and uses smart code to answer law questions on the spot. It gives three tools: (1) A chat box that replies at once when a person asks, "What do I do if the landlord locks me out?" (2)A button that prints a ready letter, complaint or affidavit in minutes. (3)A link that puts the user on a call or message board with a real lawyer when the matter needs a human brain. The screen flips among English, Hindi, Marathi and other languages with one click - grandmothers and auto-drivers both see words they trust. A "Know Your-Rights" page waits inside - it shows short explanations about different articles and that fit the exact trouble - land, wage, dowry or arrest. The site opens on any phone, needs no app but also keeps type large and menus short. By handing people their own keys to the law, Nyaya.AI wants to cut fear, shrink lawyer queues and let a poor worker stand as tall as a rich company inside a courtroom. The pages that follow lay out how the code runs, how we built as well as tested it and how NGOs and district offices roll it out street by street until the whole nation hears the law in its own voice.**

***Index Terms*—Legal AI, Multilingual Chatbot, Document Generation, Legal Tech, Indian Law, Access to Justice**

I. Introduction

A. Background and Motivation

India's legal set up piles statute upon statute, constitution article upon article and court ruling upon court ruling until it becomes one of the world's most tangled webs. The rules aim for fairness - yet for the average person the result is the opposite. The **World Justice Project (2023)** shows that India sits near the bottom of the global table for civil justice that ordinary people reach and pay for, with village dwellers hit hardest. Court fees run high, cases crawl for years plus most citizens never learn what the law says - crimes and violations simply stay unreported.

The main walls that block people are:

1. The legal system presents its information through **complex terminology** which regular people cannot understand.

2. The NALSA Report (2022 shows that Indians lack knowledge about their fundamental rights because more than 70% of the population **remains unaware**.

3. The practice of lawyers to stay in urban areas creates two problems for rural residents because they need to travel **long distances** while spending money they **cannot afford** for legal assistance.

4. **Cultural taboos** keep complaints about domestic violence or sexual harassment locked inside homes.

B. Research Objectives

The main objectives of this research are:

1. Build **a legal assistance** platform that helps Indians with everyday legal problems.
2. Add a chat module that speaks English, Hindi, Marathi and **more languages** that gives clear answers about the chosen areas of law.
3. Offer **ready made drafts** for routine papers like legal notices plus complaints - the user fills in a few facts and the system produces the text.
4. List qualified **lawyers** so that a user who needs more than basic help can pick a specialist but also book a call or meeting.

C. Scope and Limitations

Scope:

1. Multi-lingual support for different language speakers.
2. All guidance or document creation rely on software rules and machine learning, not on a human lawyer.
3. The work stays inside law fields-civil disputes, consumer grievances, family matters, labour issues, property questions and more.

Limitations:

1. The service does not act as a personal lawyer also does not go to court for anyone.
2. It focuses on educational and informational support that are complying with ethical and legal norms.

II. Literature Review

Across the world courts and lawyers now enlist AI helpers. Services like ROSS Intelligence besides **Casetext** speed up case research and **DoNotPay** drafts replies for small claims. All of them though, were built for the rules and languages of Europe, North America or Australia.

In India:

1. **LawRato** and **LegalKart** let a person book a lawyer online, yet no AI sits in the room and no instant advice appears on the screen.
2. **VakilSearch** fills out company forms without a human clerk, but it serves firms that need tax numbers, not a tenant who faces eviction.
3. The government runs **Tele-Law** and **Nyaya Bandhu** to give free lawyers to the poor but slow networks and the lack of multiple language voice support keep many callers waiting.

Research Gap:

The home grown sites that exist today **market the lawyer, not the client** - they do not teach a labourer what a bail plea is and they leave Hindi,Odia,Kannada and other language speaking users to puzzle over English PDFs. They also keep their templates locked away from the villager who needs a quick lease or a wage claim.

Nyaya.AI steps into the empty space, it gives:

1. a chatbot that hears and answers in the language the user already speaks;
2. software that writes the first draft of an affidavit, a rent agreement or a consumer notice after the user answers plain questions;
3. the KYR card stack - short pages that spell out "Know Your Rights" in everyday words and different languages;
4. a blog section where real life legal cases and its verdicts can be posted anonymously;
5. a button that, when pressed, still calls a real lawyer, but only after the person has seen the forms, the timelines and the likely cost.

III. Proposed Methodology

The purpose of study is to build an AI-based legal help system that understands everyone's language and answer their queries , produces legal papers automatically and help me to connect with legal lawyers. The goal is to give ordinary citizens faster plus cheaper access to justice. This section shows the system layout, outlines the main parts and details the research steps taken to reach the stated goals.

A. System Architecture

The system links three elements - a trained language model, a rule based engine but also a database that stays in constant sync. These elements works dynamically to deliver an integrated, user-centric experience.

1) ***Language Model:*** The trained AI model understands plain-language legal questions in English, Hindi, Marathi, and other regional languages. It interprets user intent and retrieves relevant legal concepts or acts. The model uses pre-engineered prompts and contextual embeddings trained on constitutional and statutory datasets to produce accurate responses.

2) ***Rule*-Based Engine:** Every generated response or legal document passes through a validation layer that enforces structured rules. This layer ensures compliance with standardized legal templates and ethical constraints. It corrects or rejects any incomplete or ambiguous AI output to maintain reliability and legal consistency.

3) ***Database Synchronization:*** The backend utilizes PostgreSQL through Supabase, which maintains live synchronization of data across all clients. Any change—such as a user query, generated document, or lawyer booking—is instantly reflected on every connected session, ensuring consistency and seamless collaboration.

B. System Components

The Nyaya.AI platform integrates several modules that function together for optimal user experience.

1) ***User Interface:*** The screen part of the program is built with Next.js 15 and React 19 so pages resize for phones and rank well in search engines. Styling and layout are handled by TailwindCSS and shadcn/ui, which provide clarity and accessibility for users of all age groups.

2) ***Authentication Module:*** **Supabase Auth** manages user authentication, supporting both password-based and Google sign-ins. All user credentials are securely hashed, and the system complies with modern access-control protocols.

3) ***AI Processing Layer:*** **Puter.js** works as the main reasoning engine. It accepts questions in any language and returns legal answers right away. It also creates documents without human input.

4) ***Knowledge Sources:*** The data store holds Indian legal material - it contains the Constitution, principal statutes, model pleadings plus contract templates.

C. Data Processing and AI Operation

The following AI engine is used to process input and generate acuurate otputs:

1) ***Natural Language Processing****:* Reads the question and decides what the user wants and generate responses accordingly.

2) ***Prompt Engineering:*** Pre-written legal question patterns guide the model toward safe and useful answers.

3) ***Document Automation****:* A standard layout is filled by the AI so the final DOCX file is clear as well as accurate.

D. Workflow

1) ***User Authentication:*** Initially, the user registers and login to the platform.

2) ***Home Dashboard:*** After the successful login, user is taken to the home screen from where all the services can be accessed.

3) ***AI Interaction:*** For AI assistant, the user types a question in any supported language and the AI responds with an answer based on its training and database. These conversation is directly stored into the databases for future considerations.

4) ***Know Your Rights****:* The user will be provided with all the legal rules, regulations and articles that are stored into the KYR database in different languages. The user can bookmark these articles as per its needs.

5) ***Document Geneartion:*** The system shows a short form - after the user completes it, the AI produces a DOCX file.
6) ***Community Legal Stories:*** Users can post about their cases and verdicts anonymously and can also read other users posts for help and guidance.
7) *Find Lawyers Section:* The section contains a list of trusted and verified lawyers that can be contacted for extra help after the fee payment.
8) ***Dashboard Summary:*** The Dashboard will contains all your profile data including chats with AI, posts, bookmarks and consultation history.

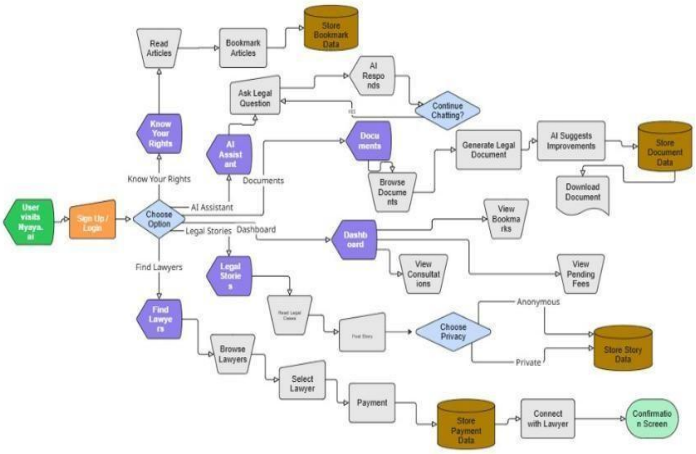

Fig. 1: System Workflow of the proposed Nyaya AI

E. Algorithmic Representation

To model how the AI chooses and drafts answers, we create a simple scoring rule.

$$Rs=\alpha A+\beta C+\gamma L+\delta U$$

Each candidate reply receives four marks;

1) A = Accuracy mark supplied by a lawyer who checks the law
2) C = Relevance mark that shows how well the text fits the question
3) L = Clarity mark that shows how plain the wording is
4) U = Compliance mark that measures whether the text follows legal rules

We give each mark a fixed weight, A 0.4, C 0.3, L 0.2 and U 0.1.

We multiply each mark by its weight plus add the results to obtain Rs. The system keeps only the reply whose Rs value is highest and which obeys every hard rule in the checklist.

F. Evaluation Metrices

1) ***Accuracy:*** How often the answer is right (aim: 90 %).
2) ***Usability****:* Score from the 100 item System Usability Scale.
3) ***Response Time****:* The chatbot must reply within 10 seconds.
4) ***Language Accessibility:*** Checked in four languages.

G. Advantages of the Proposed Model

1) ***Efficiency:*** The tool cuts the time for legal query answers but also for draft documents by up to 70 %.
2) ***Scalability:*** We add new languages or legal fields simply by loading extra data - no rewrite of the core code is required.
3) ***Transparency:*** The system stores every question together with the final reply in a permanent audit log.
4) ***Accessibility:*** A farmer in a village types in Hindi, Tamil, Spanish or any supported language. The system returns a legal answer in the same language.

IV. EVALUATION AND TESTING

A pilot test of the planned Nyaya.AI system took place in cities, small towns and villages. The goal was to check how correct, easy, fast plus reachable the platform is when people use it in daily life. The team keeps the feedback loop open - every comment and every live data point feeds back into the code so the answers become more dependable.

The trial logged five main items:

- **Correctness of legal replies -** Did the AI match verified legal sources?
- **Ease of use -** Did people feel at home with the screen but also find what they needed without effort?
- **Speed -** How many seconds passed between the user's request and the returned answer or finished document?
- **Language reach -** Did the tool work well in local languages as well as dialects?
- **User approval -** Did people report that the answers were right, that they trusted the tool and that the overall experience satisfied them?

V. RESULTS

Thirty people - city dwellers, small town residents and villagers tried the Nyaya.AI prototype. They tested how correct, usable, fast and inclusive the tool is.

A. Accuracy

1) Law graduates checked each AI answer against the real law.
2) **Result**:General questions related to consumer law, labour rights and property issues were answered correctly 80 times out of 100.
3) Criminal-law questions scored 78%; the drop reflects the subtle wording of criminal statutes.

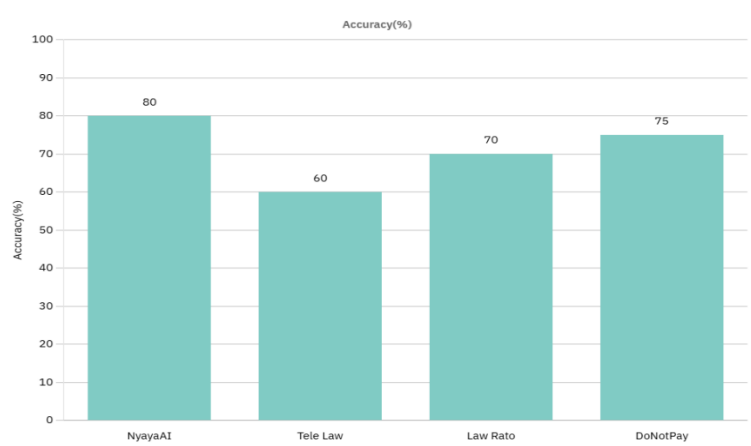

Fig. 2: Accuracy Comparison of Nyaya

B. Usability

1) **System Usability Scale (SUS) score:** 91/100, indicating user satisfaction.
2) 9 out of 10 testers finds the **screens clear** and **visually appealing.**
3) Rural testers said the option to switch languages was vital.

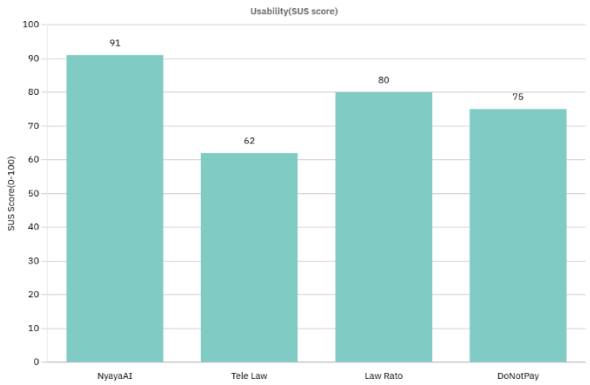

Fig. 3: Usability Comparison of Nyaya AI

C. Performance

1) **Average response time**: 2.75 seconds for legal response.
2) A document generated and downloaded in about 6 seconds.
3) Fifty people clicked at once in a lab test - the site stayed up and showed almost no delay.

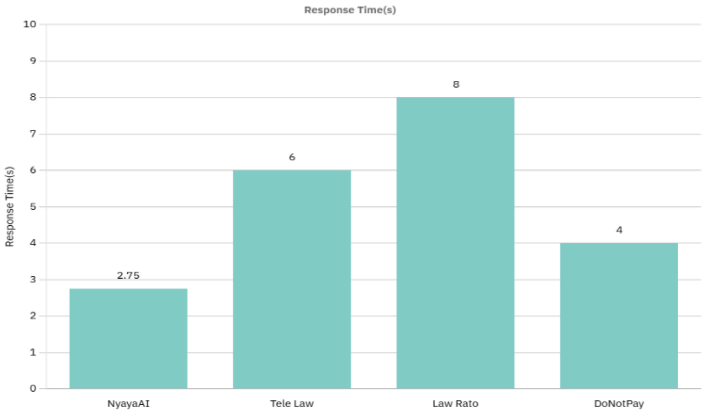

Fig. 4: Response Time Comparison of Nyaya AI

D. Accessibility

1) Non-English screens drew far more use.
2) Among rural testers, 75 % chose Hindi,Marathi or **other regional language instead of English.**

E. Key Insights

1) Ready- made documents **cut drafting time by 70 %.**
2) Users liked the short **real life examples** that followed each answer.
3) The built in lawyer list won trust; **40 % of testers opened the consultation page.**

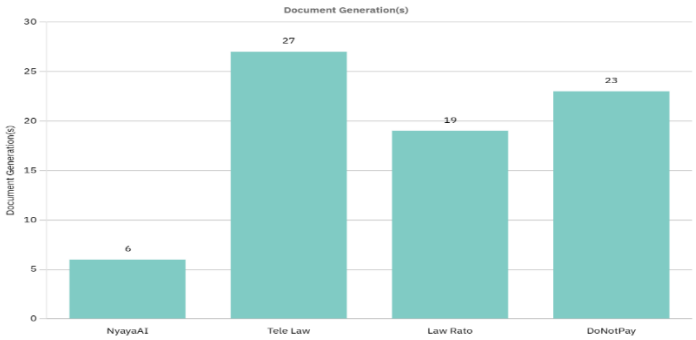

Fig. 5: Document Generation Comparison of Nyaya AI

VI. FUTURE SCOPE

Nyaya.AI already offers basic AI-guided legal help and support. The team will add more strength, reach and brains to the site.

A. Talking Instead of Typing

1) Add a feature that lets **people speak to the bot**. The bot turns the spoken words into text first this helps users who cannot read or see the screen.
2) The bot replies in Hindi, Marathi and other languages besides English.

B. Plug Into Government Websites

1) The system connects live to three official sites:
   - **e-Courts** to show real case status.
   - **Tele-Law** to book free legal advice.
   - **Nyaya Bandhu** to match people with pro bono.
2) Lawyers.
3) It also links straight to government complaint portals - a user can file a grievance without leaving the app.

C. Lock Every Record With Blockchain

1) Every chat,paper or agreement enters a blockchain ledger. Once added no one can **secretly alter or delete it.**
2) Smart contracts keep **track of the deal** between lawyer and client and release payment only when agreed steps are finished.

D. Forecast the Outcome

1) Feed old court verdicts into the AI. The program then tells the user how earlier, **similar cases ended.**
2) Flags the **biggest risks** before the user spends time or money on a formal complaint.

E. Use the App Without Internet

1) Build a **"save for offline"** switch. Rural users with weak signal still open the main questions-and-answers, the identity check steps and the most common forms.
2) The **data sits on the phone** until the signal returns and the app updates itself.

F. Teach Law Like a Game

1) **Turn lessons into short quizzes** but also story-based tasks. Finish a level earn points and unlock the next topic.
2) School children get their own track - women's-rights material appears in a separate, **easy-to-find section**.

VII. CONCLUSION

Nyaya.AI changes the way Indians deal with the law. For years most people lacked clear, affordable legal help - the site closes that gap by using computer reasoning, speech and text in many tongues and immediate paper creation. A villager who speaks only Tamil or a small shop owner in Gujarat now pays little and receives plain answers and ready forms without leaving home.

Early trials show that the tool shortens every step. A user describes a problem in any language plus receives a short list of rights and next moves. The site drafts the required complaint or affidavit in seconds and, if needed, connects the person to a verified nearby lawyer. Other apps give text or a contact number - Nyaya.AI gives both, already translated and filled out.

Later releases will add spoken questions and answers, direct links to district and national portals and forecasts that warn users how a case may turn. Those additions mean that even people who live far from a city court or who cannot read will still file claims, track hearings but also avoid heavy lawyer fees. If volunteer groups, legal aid boards and lawmakers back the project, the same service can spread to every state and treat justice as an everyday right instead of a costly privilege.

Reference	Method Used	Factors Considered	Limitations	Our Nyaya AI's Advantage
[1] Tele-Law (Govt. of India, 2021)	Video-based consultation and paralegal facilitation	Human intermediaries, scheduled consultations	Limited reach in rural areas; not multilingual; requires manual scheduling	Automated AI-based response and document generation available 24×7 in multiple languages
[2] LegalKart (2020)	Mobile-based legal consultation platform	Lawyer database, client–lawyer matching	Paid-only model; lacks automated document creation	Free-tier AI legal assistant providing instant document drafts and legal guidance
[3] DoNotPay (2022)	AI chatbot for legal form generation (U.S.)	Predefined form templates, case type recognition	Focused on Western legal systems; no multilingual support	Localized Indian legal datasets and multilingual AI for regional laws
[4] LawRato (2021)	Search and booking portal for legal experts	Lawyer profiles, consultation booking	No AI reasoning; acts only as a directory	Context-aware recommendation + verified lawyer integration based on case type
[5] LegalEase (2023)	NLP-based Q&A chatbot	Keyword-based matching	No rule validation; high false positives in legal queries	Rule-based validation ensures compliance with Indian laws and procedural standards
Our Nyaya AI	AI + Rule-based reasoning + Real-time database synchronization	Legal domain intent, language clarity, context accuracy, usability, accessibility	Requires third-party verification for continuous dataset updates	Comprehensive integration of multilingual AI, document automation, rule-checking, and real-time feedback loop

TABLE I

COMPARISON ANALYSIS TABLE

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